

Intracellular Water, Metabolism and Cell Architecture: Part 2

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INTRODUCTION

Biologists seem to pay little attention to physical theories that attempt to explain living systems without detailed reference to the elaborate structure that we know exists, notably at the cellular and subcellular levels. I suspect this explains, at least to some extent, why Fröhlich's theory of coherent excitations (See Fröhlich, 1980) has almost gone unnoticed by the biological community. Therefore, the present chapter will attempt to describe what we know about certain features of the structure and function of eucaryotic cells, selecting those aspects that seem to be most relevant to the theory.

The title carries "part 2" because an initial attempt in that direction was made three years ago (Clegg, 1981a). At that time it was pointed out that an intricate network of protein structures known as the "cytoskeleton", and notably the then very recently discovered microtrabecular lattice, extends from the cell surface throughout the cytoplasm. Evidence was presented that suggested that this architecture might be intimately associated with the metabolic machinery of cells and that it also determined the physical properties of a large fraction of intracellular water by virtue of its vast surface area. I proposed that all these relationships could play a significant role in the establishment and propagation of the coherence predicted by Fröhlich's theory. In the following section I will reconsider the essential features of the ideas presented in part 1 (Clegg, 1981a) and evaluate them in the light of research performed since that time. It should be stressed that space limitations prevent anything approaching a review of these very active research areas. Only recent reviews and key articles that provide access to the literature can be cited. As pointed out in the previous attempt (Clegg, 1981a) I claim no expertise in phy-

sics in general, or in coherent phenomena in particular, and am in no position to support or refute the theory under consideration here. My objective is to provide a cellular point of view.

CELLULAR ARCHITECTURE

In the present context the group of cellular structures that appear to merit closest attention are the cytoskeletal components: microtubules (about 250Å diameter) made from the protein tubulin; microfilaments (about 60Å diameter) made from the protein actin; intermediate filaments (about 100Å diameter) whose protein composition varies. These and related structures have been the object of an enormous amount of research, much of which has recently been considered in the two published volumes of a Cold Spring Harbor Symposium (Albrecht-Buehler, 1982). In addition to these, the work of Porter and associates has indicated the existence of a highly branched three dimensional network that not only extends throughout the cytoplasm but is believed to surround virtually all cytoplasmic structures with the possible exception of mitochondria (see Porter and McNiven, 1982). This structure, abbreviated "MTL", contains a large amount of actin which appears to be its major structural protein; however, a wide variety of different proteins are also closely associated with it (see Schliwa, et al. 1981) a point to which I will return. The relationship between actin microfilaments and the MTL is controversial and I should point out that alternative views of the actin networks in cells have been presented (see Heuser & Kirschner, 1980; Albrecht-Buehler, 1982). My position is that Porter and colleagues have made a very strong case for the reality of the MTL, and I proceed here on that basis.

Of great importance to this discussion are extensive connections made between plasma membrane and the actin networks (see Poste and Nicolson, 1981; Avnur & Geiger, 1981; Penman et al. 1982). It is well documented that many extracellular signals, a number of hormones for example, exert their influence by first binding to receptors at the cell surface. This binding is followed by cytoskeleton - mediated events and/or the production of "second messengers" (cAMP, cGMP) via activation of the appropriate cyclase enzyme, also in or near the cell membrane. Such "transmembrane modulation" plays a dominant role in cellular function. Because the MTL makes direct or indirect connections with these cell surface receptors as well as nearly all other cytoplasmic structures this system provides an excellent means of intracellular communication.

But the situation is made even more interesting by the detailed work of Penman and associates (see Penman et al. 1982 for review). They demonstrated that the functional units of protein synthesis, polysomes, are also contained within or on the MTL (those polysomes that are not attached to endoplasmic reticulum). Consequently, these key cell components are structurally connected with cell surface receptors, as well as other cell structures, via the MTL. Such connections clearly must have important functional consequences. The Penman research group has also recently shown that connections exist between nuclear chromosomes, the nuclear cytoskeleton and the MTL (Capo et al. 1982). Those observations are of potentially enormous importance because they allow for the possibility of genetic control, and other nuclear events such as those involved with cell division, all potentially mediated via the actin networks.

The picture that emerges from this brief description (diagrammed in Fig. 1) is one in which essentially all of the ultrastructurally recognizable structures in cells are interconnected by the MTL. Dotted lines represent the MTL in association with a single cell surface receptor. However, it should be realized that a very large number of these receptors exist and, although direct evidence is lacking, it is possible that most of these are at least indirectly associated with the MTL. Relationships between enzymes and the MTL will be considered later in this paper.

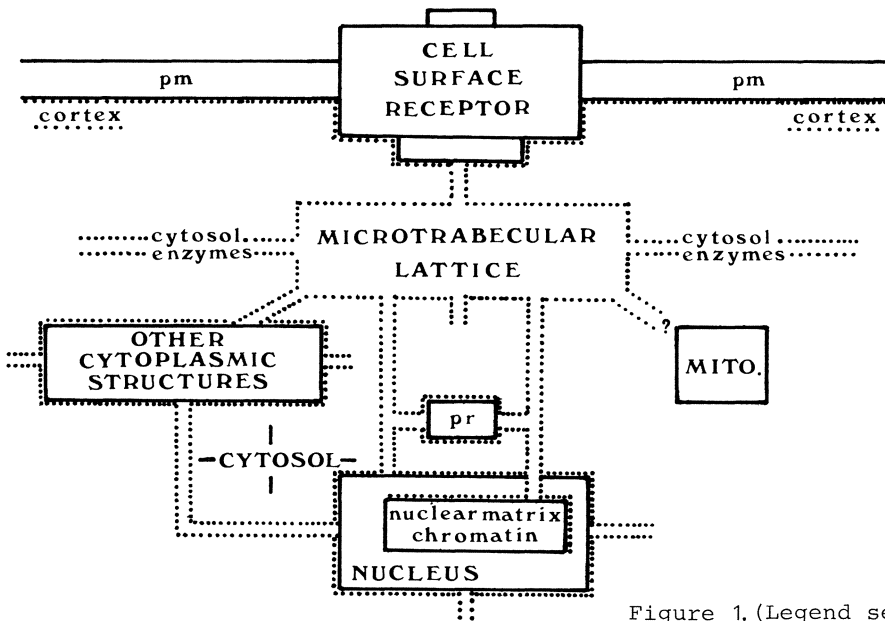


Figure 1. (Legend see p. 165)

An analogy, perhaps too fanciful, would describe the cell as an electronic schematic diagram in which the various electronic components (cell organelles, etc.) are interconnected and controlled by inputs via wiring (actin networks, MTL) from other parts of the instrument (cell) including major controlling switches (cell surface and intracellular receptors). Input from one small area (the cell surface, for example) can be amplified enormously and rapidly conveyed to the appropriate part of the cell.

Before leaving architectural matters it is appropriate to briefly consider the dynamics of this system. I concentrate here on the actin-based components. Many studies have been carried out on the "turnover" of actin structures in vitro (see Korn, 1982) and in vivo (see Kreis and Birchmeier, 1982). In the test tube single molecules of globular actin (G-actin) can, under suitable conditions, be polymerized into filaments (F-actin). The growth of such filaments takes place at the two ends, one being faster than the other. The net rate of growth in vitro can be as fast as 90 G-actin monomers added per second per filament, amounting to an increase in length of about $3000\overset{\text{O}}{\text{\AA}}$ ($0.3\text{ }\mu\text{m}$) per second per filament. Recent work has clearly shown that the turnover of the actin networks in living cells is constant and also rapid, although precise numbers are apparently not available.

The energetics of polymerization is of some interest to us (see Korn, 1982). Each actin monomer (G-actin) contains 1 molecule of bound ATP, the free energy currency in living cells. It is a puzzling fact that this ATP does not provide the free energy for polymerization; however, it is split into ADP (which remains bound) and inorganic phosphate (which is released) after F-actin is formed. Thus, the formation of actin filaments involves considerable free energy input, well in excess of that required for polymerization. Although several proposals have been made, the role of this energy expenditure remains obscure. In the final section of this chapter I will speculate on how the energetics of actin polymerization might be related to Fröhlich's theory.

Figure 2 summarizes several features of the foregoing discussion and provides some additional aspects of cell architecture, admittedly oversimplified and not entirely accurate.

Figure 1. Diagrammatic Representation of the Microtrabecular Lattice and its relationship to other Cell Structures (pm = plasma membrane, pr = polysomes; "cytosol" refers to the aqueous compartments of the cytoplasm)

Figure 2A represents an actin filament associated with the plasma membrane. These associations are extensive and probably involve specific linkage protein (LP) such as vinculin (Avnur & Geiger, 1981; Isenberg et al. 1982) or several others (see Albrecht-Buehler, 1982) connected to an integral membrane protein (IP). It should be noted that a wide variety of specific surface receptors have been shown to be associated with actin networks, both structurally and functionally (for example, Hunt and Hood, 1982; Snyderman and Goetzel, 1981; Conti-Tronconi, 1982; Puca and Sica, 1981; Prives et al. 1982; Pribluda and Rotman, 1982). Thus, the integral protein (IP) in Fig. 2A can also be viewed as such.

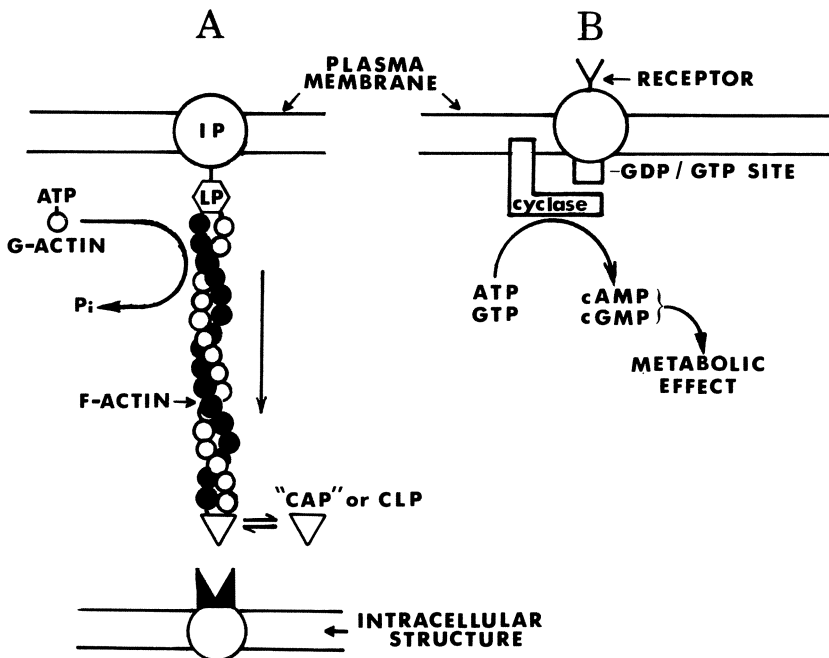


Figure 2. Diagram of Actin Filaments (A) and the Transmembrane Control of Cyclic Nucleotide Production (B)

There is some evidence that filament growth occurs by addition of G-actin to the membrane-associated end of the filament (see DeRosier and Tilney, 1982). Many proteins have been implicated in the control of filament growth and connection to other cell parts represented (not very precisely) as "cap" and cross-linking proteins (CLP). These affect filament length and the organization of three dimensional networks. The magnitude of this activity is considerable since actin can make up as much as 15-20% of total cell protein.

Figure 2B depicts the transmembrane modulation process discussed previously in this section. This complex mechanism is turned on or off depending upon the presence or absence of specific ligand bound to the receptor. The cyclase enzyme (adenyl or guanyl) when activated produce the cyclic nucleotides (second messengers). These systems also require significant energy input when active: for each mole of cyclic nucleotide produced 3 "high energy" phosphoryl groups are utilized, one of which involves GTP at a regulatory site (Fig. 2B). It is of some interest in the context of this discussion that the cytoskeletal components have very recently been implicated in the activities of these systems as well (Sahyoun et al. 1981; Cooper, 1982; Rasenick, 1981; Westermarck and Porter, 1982; Browne et al. 1982). That feature is not shown in Fig. 2B. I will return to these important matters in the last section of this paper.

ENZYMES AND CELLULAR ARCHITECTURE

It has long been known that some enzymes in cells are integral parts of various membrane systems or other structures such as ribosomes and glycogen granules. However, most of intermediary metabolism has traditionally been considered to result from enzymes freely dissolved in solution within the aqueous compartments of cells, the "soluble enzymes". That conception, based largely on cell disruption studies, is currently being questioned and reexamined because of the increasing evidence that few enzymes exist free in solution when intact cells are examined by appropriate techniques. Access to the extensive literature on this and related matters is available through recent reviews (Kempner, 1980; Masters, 1981; Clegg, 1981a,b; Fulton, 1982) and papers (Weber and Bernhard, 1982; Klyachko et al. 1982; Morton, et al. 1982; Clegg, 1982a; Mansell & Clegg, 1983). I believe that an examination of this evidence clearly indicates that the great majority of intracellular macromolecules, including enzymes are probably not free in solution, but are instead loosely associated with formed structure. The question of their location is of some significance, but remains partially unanswered. In skeletal muscle tissue there is very good evidence for the association of most glycolytic enzymes with the thin filaments, which are chiefly actin (see Masters, 1981; Morton et al. 1982). Moreover, a recent study by Schliwa et al. (1981) strongly suggests that many cytoplasmic proteins are loosely associated with the actin networks of non-muscle cells. That paper, which combined ultrastructural studies with an analysis of cellular proteins in normal and detergent-treated cells, in conjunction with evidence just cited, provides good reason to postulate the extensive organization of enzymes along (or with-

in) actin networks of cells. It is quite likely, furthermore, that enzymes of a given metabolic pathway are organized into loosely associated multienzyme complexes. That is, these enzymes may be in close proximity to each other as well as being associated with cell architecture. The basis for that possibility comes from a number of studies on metabolic flux measurements, calculations of metabolite-transit times, and kinetic tracer studies of intracellular pools (see Kempner, 1980; Masters 1981; and references therein).

Figure 3 shows a diagrammatic representation of the traditional (Fig. 3B) and "organized" (Fig. 3C) conception of the disposition of enzymes in the aqueous compartments of cells.

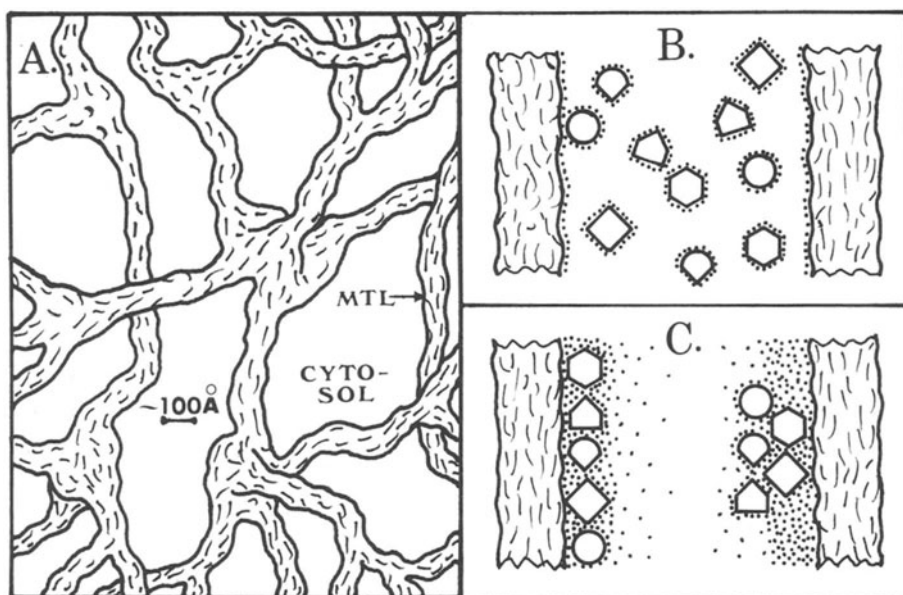


Figure 3. Diagrammatic Representation of the Microtrabecular Lattice (A) and its Possible Relationships to Cytosol Enzymes (B,C)

My own bias leads me to use the MTL (shown in Fig. 3A) as the major architectural site for such organization. Its variable thickness (50-200 Å) is consistent with the description in Fig. 3C. Of course, locations other than actin networks could also be involved. The advantages of such organization, compared with a random solution-based metabolism, have frequently been pointed out (see Masters, 1981).

At this point, then, a reasonable case has been made for the existence of continuity of structure from cell surface throughout the cyto-

plasm and extending into the nucleus. If the view represented in Fig. 3C is correct the enzymes of intermediary metabolism, by virtue of their self-association (for a given pathway) and their close relationship to this architecture, represent a very high degree of organization. It seems possible that the entire metabolic machinery of eucaryotic cells is involved in this organization. In fact, there is reason to believe that even many small molecules (metabolites) may also be associated with macromolecules and/or cell structures, perhaps existing in "free and bound" forms. That is certainly true for the important coenzymes NAD^+ and NADP^+ and their reduced forms, and Wheatley & Inglis (1981) have presented and reviewed extensive evidence for amino acid binding in cells. Finally, Kempner (1980) has described experiments that show the nonrandom distribution of fluorescein (MW about 350) in the aqueous compartments of intact cells. The compartmentation of small metabolites as well as enzymes is a topic of much current interest. This compartmentation has been described as "structural" (metabolites segregated by membranes) and "dynamic" (metabolites segregated by functional means). Current views are available in publications by Hess et al. (1980 and Sies (1982). It certainly appears that the molecular organization of cells is far more extensive than commonly believed.

The foregoing discussion leads us to a consideration of the aqueous compartments: those regions of cells in between the vast and complex architecture observable by microscopy.

INTRACELLULAR WATER AND THE AQUEOUS COMPARTMENTS

These compartments can be considered as those regions of eucaryotic cells which contain no demonstrable ultrastructure. Obviously, our understanding of these compartments has changed as more and more structure was demonstrated to exist. Thus, about 20 years ago the cytosol would inadvertently have included all cytoskeletal components as well as the MTL, simply because those structures were not known to exist. Therefore, our current view of the cytosol must consider the possible existence of organization which is not observable by current methods. Given this qualification, the cytosol would seem to be a relatively dilute aqueous solution containing only inorganic ions, small metabolites and, perhaps, a few macromolecules. That picture is much different from the traditional view (Fig. 3B). A hazy conceptual boundary exists between cytosol and the bordering non-aqueous ultrastructure because the system is dynamic, and the composition of the cytosol is likely to be constantly changing. A similar description can be given for the aqueous compartment of the nucleus, the "nucleoplasm."

Water dominates the composition of all these compartments of course, being, on a mole fraction basis, over 0.90 for all cell constituents. It is necessary to realize that this water is exposed to the truly vast surface area presented by cellular architecture. An example from the very recent work of Gershon et al. (1982) is illustrative. Using modern image analysis methods they estimated the surface area and volume of the MTL and associated cytoskeletal components in mammalian tissue culture cells. For a single cell of 16 μm diameter the surface area of these cytoplasmic structures is between 47,000 and 95,000 μm^2 . This calculation excludes the nucleus which is taken to be 10 μm diameter. Thus, the total cytoplasmic volume is 75% of the total cell volume, or about 1620 μm^3 . Therefore, the surface to volume ratio of the cytoplasm is about 30 to 60, and this does not consider the surface area of all of the other cytoplasmic architecture (membranes, granules, ribosomes etc.) which is very considerable (see Clegg & Drost-Hansen, 1978). The nucleus has its own "cytoskeletal" system which is also enormous (see Penman et al. 1982).

On this basis alone we should expect a large fraction of cell water to be in reasonably close proximity to ultrastructural surfaces (Fig. 3A). That is significant because water within 25-30 $\overset{\circ}{\text{A}}$ of biological surfaces (see Lis et al. 1982) as well as non-biological ones (see Israelachvilli, 1981) is known to exhibit physical properties that differ significantly from those of pure water. Such "vicinal water" (Drost-Hansen 1978) is depicted by stippling in Fig. 3C.

Indeed, 25-30 $\overset{\circ}{\text{A}}$ is a minimal distance over which the influence of the surface on water is "felt". Drost-Hansen (1978) has compiled evidence that this influence can extend over 500 $\overset{\circ}{\text{A}}$ from such surfaces. If correct, then all cell water will be altered (see Fig. 3A). However, the traditional view of intracellular water has been that practically all of it has properties that are identical to those of pure water (reviewed by Clegg, 1982b). Indeed, much of current thought about cell structure and function is based on that description of intracellular water. In contrast, an increasing body of recent evidence strongly indicates that a large fraction of the water in living cells exhibits properties that are distinctly different from those of the pure liquid. Among the most compelling to me are the studies by Horowitz, Paine and associates (see Paine & Horowitz 1980; Horowitz and Pearson, 1981) who used an ingenious technique to show that the solvent properties of cytoplasmic water in amphibian oocytes differ profoundly from those of pure water. They argued that their results are probably valid for animal cells in general. It is very difficult to reconcile such results with the traditional view. These and other descriptions of cell water,

including the traditional one, can be found in books edited by Keith (1979), Drost-Hansen & Clegg (1979) and Franks (1982).

I believe we are approaching at least a partial resolution of this long-standing controversy, and it is my view that the traditional position will undergo drastic modification. It is even possible that none of cell water has the properties of pure water, a view that has long been advocated by Ling (1980) and strongly supported by Negendank (1982) and Hazlewood and associates (see Hazlewood, 1979). All non-traditional views on cell water invoke the influence of intracellular surfaces. Thus, there is a sound basis for the perturbation of a large fraction of the water in living cells.

The significance of altered cellular water, compared with the pure liquid, can now be considered. If the view that virtually all metabolism occurs at or near the surfaces of cellular structures is accepted then it is evident that the enzymes that perform this role find themselves in a microenvironment that is clearly not a dilute aqueous solution. It is well known that the microenvironment can have a profound influence on the kinetic, thermodynamic and regulatory properties of enzymes (see Masters, 1981) and particularly the thorough and quantitative analysis by Welch and Keleti (1981) and Welch et al. (1982). Careri (1982) has given an impressive account of the importance of hydration to enzyme dynamics (also see Careri et al. 1979). These treatments leave little doubt that descriptions of cell metabolism and its regulation that are based on results from dilute solution studies in vitro are likely to be incomplete at best, and perhaps largely incorrect. The forces involved in enzyme-enzyme and enzyme-structure associations of the type postulated here (Fig. 3C) are not well understood (see Masters, 1981) although it is clear that they are not covalent. Because changes in water structure usually take place upon association of molecular sub-units into more complex aggregates, as in hydrophobic interactions for example, it is possible that water might participate similarly in the proposed enzyme-enzyme and enzyme-cell structure associations (Clegg, 1981a & 1982a,b). That possibility requires that the water proximal to both the ultrastructural surface and the enzyme has lower entropy (more "structure") than the water that is more distant from such surfaces. Association of the two might then be driven by the entropy increase attending the release of the more "structured water". Of course, other interactions are likely to be involved.

The description of cell structure and function given in the preceding sections of this chapter will now be considered in the context of Fröhlich's theory.

ON A CELLULAR BASIS FOR FRÖHLICH'S THEORY OF COHERENT EXCITATIONS IN BIOLOGICAL SYSTEMS

As I understand it, Fröhlich (1980) has proposed that strong electric fields ($\approx 10^7$ volt/m) in the cell membrane lead to strong electric polarization of membrane-bound molecules, notably proteins, which oscillate coherently leading to electric polarization waves. A certain minimum rate of "metabolic energy input" is required to establish that coherent behavior. To my knowledge the nature and location of energy input have not been clearly specified, although the cell membrane is obviously involved in this respect. From all that we know about cellular energetics, energy transactions almost always involve the removal of phosphoryl groups from ATP (or related ribonucleoside triphosphates). It would seem that the actin-networks in cells might be intimately related to the metabolic energy input required by the theory by virtue of their close proximity to cell membranes, their generally constant turnover, and the involvement of ATP in their formation (Fig. 2A). Recall that G-actin bound ATP is split after the actin monomer becomes part of the filament. Because this free energy transaction occurs after, and is not lost during polymerization this might provide a means of channeling the energy input required for coherence which arises due to "the sharing of energy by many macromolecules". Sufficient reason has been given in this paper to believe that the actin networks provide an excellent ultrastructural coupling device, a means of "storing energy" and propagating coherence throughout the cell.

Another ultrastructural feature of relevance is the ability of cells to control the rate of G-actin addition; thus, turnover is regulated and the system can be "switched" to higher or lower levels. Actin monomers appear to be added chiefly at nucleating sites near the cell surface (plasma membrane) to bring about filament growth, although there is not uniform agreement on that point. Since these sites often appear to be very near or directly connected to specific receptors in the plasma membrane (and/or transmembrane elements associated with them) very weak stimuli impinging on the cell surface could rapidly be transmitted throughout the cell (Fig. 2A,B). Such a system also allows for very large amplification so it is reasonable to speculate that this mechanism could be involved in the "non-thermal" microwave effects predicted by Fröhlich's theory. Such effects, although the subject of some debate, can either be positive or negative - that is, they stimulate or inhibit biological processes (or can have no effect). That is readily accounted for since the cell surface receptors could be activated or deactivated, depending on their status at the time of irradiation

(Fig. 2). Significant cellular effects could, therefore, result from very low levels of energy input. Moreover, if the proposed relationship between enzymes, the actin network and their connections is correct (Figs. 2,3c) then there is a direct link between such "sensitive domains" and the fundamental metabolic machinery from which so much of cellular structure and function originates. More on this later.

The involvement of cellular architecture has also been considered by Del Giudice et al. (1981, 1982a) but from a much more physical viewpoint, and within the context of Fröhlich's theory. Taken together with the more "biological" framework outlined here the foregoing speculation seems perhaps more reasonable. In addition, Berry (1981) has proposed an interesting electrochemical "model" of metabolism that shares several features in common with the Del Giudice group as well as my own views. Although Berry was apparently not aware of Fröhlich's theory, his model seems to be to be compatible with it.

Another feature of the theory under consideration concerns the possibility of frequency-selective long range interactions which arise from coherent excitation. Thus, an interesting possibility presents itself: the association of enzymes with the actin networks could involve their initial attraction by such attractive forces. Of some interest in this connection is the work of Starlinger (1967) who found that electrical stimulation of rat muscle in vivo increased the amount of the enzyme aldolase that was bound to the muscle structure. The effect "decayed" in less than 10 minutes post-stimulation. More recently, Clarke et al. (1980) have clearly shown that electrical stimulation increases the binding of most glycolytic enzymes to the actin filaments of bovine muscle. They offer several possible explanations, but point out that the mechanism of this electrically-induced enzyme binding is unknown. The Fröhlich effect could conceivably be involved. The theory also predicts that substrates will be attracted to their enzymes, provided that the former possess a vibrational frequency close to that of the coherent modes of the metastable enzyme. These possibilities will probably dismay the biochemist who is inclined to think only in terms of very short range interactions between enzymes, their regulators and substrates. As admitted earlier, I am in no position to critically judge Fröhlich's theory, but this single example indicates its potentially profound importance to metabolism and its control, should it be shown to be valid. One point seems clear: metabolic organization in intact cells is far far greater than previously believed.

A final consideration concerns the involvement of cellular water. I have presented a brief account of what I believe to be compelling

evidence that a large fraction of intracellular water is sufficiently close to cellular architecture as to undergo changes in its physical properties compared with water in dilute aqueous solution. This "perturbed" water could play an important role in Fröhlich's theory. I previously suggested that the water within about $30\text{\AA}$ from enzymes and ultrastructural surfaces might be involved in slowing the thermalization of energy released by the postulated coherently oscillating macromolecules and speculated on its possible participation in "non-thermal" microwave effects (Clegg, 1981a). Del Guidice et al (1982b), accepting the existence of the abundance of such interfacial water, have proposed that coherence among water dipoles should arise, depressing their usual thermal motion and forcing them to oscillate in phase. They go on to suggest that resonance between macromolecular dipoles ("excited by collective motion") and water dipoles surrounding the macromolecule may be involved in altering the properties of that surface water. It is not too unreasonable to propose that water in the vicinity of enzyme-actin network complexes must play a decisive role in metabolism and its regulation even though we only begin to dimly perceive the details of its involvement. Should Fröhlich's theory be proven correct it is clear that cellular water must be involved in a major way (Clegg, 1981).

CONCLUDING REMARKS

I am well aware of the fact that this chapter has contained a large amount of speculation. To some extent that speculation is magnified by space limitations which prevent thorough documentation of positions taken that are not in accord with current conventional wisdom, notably with regard to enzyme organization and the properties of cell water. On those two accounts I will stand by the views summarized here because I believe the evidence in their support is more than adequate. With regard to my interpretation of these views in terms of Fröhlich's theory I am much less confident. But it is fair to say that any attempt to deal with new and unusual ideas will most likely not result in a rigorous or even correct treatment. It appears that the idea of coherence in biological systems is quite new, at least to most biologists. As William James said: "When a thing was new people said - 'It is not true.' Later, when its truth became obvious, people said - 'Anyway, it is not important.' And when its importance could not be denied, people said - 'Anyway, it is not new.'" If we view these as three stages of scientific discovery it would appear to me that coherent excitations in living systems currently reside in the initial stage, or perhaps just beyond.

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